Macroevolution, Assembly Audits, Telomeric Divergence, and Pleiotropic Rate Constraints Shaping Lifespan, Brain Size, and Cancer Vulnerability in Parrots

Angelo

Centro Universitário do Planalto Central Apparecido dos Santos (UNICEPLAC), Brasília, DF, Brazil

Reproducible repository: `github.com/redkk123/Parrots` (`claude/new-session-lwh7xh`)

November 10, 2026 — Complete Pre-Registered Study, 71-Species Extension, 3D Structural Mapping & 9-Point Red-Team Statistical Audit

Abstract

Abstract. Parrots (order Psittaciformes) combine extreme longevity (30 to $>$ 60 years in macaws, cockatoos, Amazons, and *Strigops habroptila*) and primate-like encephalization with short-lived, neoplasia-prone lineages such as the budgerigar (*Melopsittacus undulatus*). Under a pre-registered three-phase protocol, we combined comparative neoplasia epidemiology (15,881 necropsies across AZA and UC Davis databases), phylogenetic comparative allometry and Ornstein-Uhlenbeck regime-shift modeling across two topologies (Burgio et al. 2019 supertree, $n = 413$; Smith et al. 2024 molecular phylogeny, $n = 399$), comprehensive quality and taxonomic verification of 126 genome assemblies (`compleasm aves_odb10`, Mash nuclear divergence, and mitogenome BLAST), branch-site and selection-intensity tests across a 101-gene panel (modules M1–M8, 58-species discovery + 71-species confirmatory extension, benchmarked against 298 control genes and raw SRA reads), AlphaFold 3D domain mapping, and genome-wide relative evolutionary rate scans (`RERconverge`, 7,756 genes, 1,000 Brownian permulations) subjected to a nine-point adversarial "Red-Team" audit. Macroevolutionary analyses support a long-lived ancestral regime (root optimum +0.22) followed by a topology-sensitive lifespan reduction in Psittaculidae ($p = 0.003$ Burgio vs. $p = 0.050$–0.052 Smith, $\lambda = 0.88$), while veterinary necropsies pinpoint extreme tumor incidence uniquely to *M. undulatus* (20.8% AZA / 25.9% age-standardized UC Davis vs. 1.5–5.4% in lories, OR $= 17, p \leq 10^{-16}$). Auditing 126 assemblies uncovered cross-species mitogenome contamination in 61/96 Iridian short-read genomes and four severe nuclear swaps/chimeras, yielding 71 validated species (BUSCO $\geq 70\%$, zero technical bias with longevity, $p \geq 0.11$). Across tumor-suppressor modules (H8), we refute gene loss or relaxed selection: all 329 initial disrupted calls in 58 species (451 in 71 species) proved to be reference-divergence, heterozygous, or high-GC exon-1 coverage dropouts, and no module deviated from the 298-gene null after correction. Standard `aBSREL` flagged episodic divergence in 5/8 (62.5%) telomeric M7 genes (`RTEL1`, `TERF2`, `TERT`, `ACD`, `TERF1`) plus `MYC` on *M. undulatus* (vs. 0/19 FDR in control genes, Fisher $p = 0.00069$), confirmed as 100% orthologous via *S. habroptila* dual-reference coordinates; however, window cleaning removes `RTEL1`, `ACD`, and `TERF2`, and `Double+Triple` multinucleotide models absorb all five ($p = 0.967$–1.00), implicating clustered C-terminal (`RTEL1`) and OB-domain (`POT1`) multinucleotide dynamics rather than dispersed positive selection. Genome-wide (H6), unadjusted longevity- and brain-associated `RER` profiles correlate at $r = 0.666$–0.667, yet our Red-Team audit reveals that true OLS/branch orthogonalization halves residual brain coupling ($r = 0.332$–0.389 vs. 0.639 from unwhitened PGLS residuals), phenotypic covariance depends heavily on Psittaculidae (r_{tip} drops from 0.478 to 0.195), and 89.3% (117/131) of shared candidate hits (`PTEN`, `MAD2L2`, `HSPA8`, `CTNNB1`) have 97–99% zero-length branches mirroring the genomic background vector $\pm\sqrt{\text{nv}}$. Partitioning hits into Tier A (ultra-conserved background-coupled) and Tier B (14 informative genes with ≥ 20 nonzero branches, led by `TKT`, `PHOSPHO1`, `SLC4A7`, and `HUS1`) establishes a rigorous benchmark for avian comparative genomics.

Keywords: Psittaciformes · Peto's Paradox · Longevity · Encephalization · Telomeres (`RTEL1`, `TERT`) · `PTEN` · `RERconverge` · Statistical Audit

1 Introduction

Understanding how large, long-lived, and highly encephalized vertebrates suppress somatic malignancy over multi-decade lifespans remains a foundational question at the intersection of evolutionary biology, oncology, and neuroscience (Peto's paradox). Among endothermic vertebrates, parrots (order Psittaciformes) occupy an exceptional evolutionary space. Despite elevated avian metabolic rates and high plasma glucose, macaws (*Ara*, *Anodorhynchus*), cockatoos (*Cacatua*, *Probosciger*), Amazons (*Amazona*), and the flightless kakapo (*Strigops habroptila*) regularly survive 30 to over 60 years while evolving telencephalic neuron counts comparable to primates. Conversely, within the Old World family Psittaculidae, small species such as the budgerigar (*Melopsittacus undulatus*) exhibit brief lifespans and extraordinarily high frequencies of spontaneous neoplasia in veterinary pathology series.

Prior genomic studies of parrot longevity and cognition relied primarily on pairwise comparisons of a few long-lived taxa against non-parrot outgroups (e.g., Wirthlin et al. 2018). Yet order-wide comparative genomics has recently been reshaped by hundreds of low-cost short-read draft assemblies deposited into public repositories (such as the Iridian Genomes project; Hains et al. 2022), alongside two competing phylogenies (Burgio et al. 2019 supertree vs. Smith et al. 2024 sequence-based phylogeny). Without systematic assembly auditing, short-read fragmentation, cross-sample contamination, and high-GC

exon dropout can manufacture false signals of gene loss or accelerated molecular evolution, while uncalibrated macroevolutionary rate correlations can mistake shared clade structure or genome-background normalization artifacts for independent gene-level convergence.

Here we report the complete execution of a pre-registered three-phase investigation (`results/phase0/analysis_plan.md`), augmented with AlphaFold 3D structural mapping, a 71-species confirmatory panel extension, and an exhaustive nine-point Red-Team statistical audit designed explicitly to falsify our own primary claims.

2 Materials and Methods

2.1 Phase 0 & 0b: Phylogenies, Traits, Regime Shifts, and Necropsy Epidemiology

Life-history data were curated from Smeele et al. (2022; BaSTA Gompertz captive life expectancy at birth, $n = 218$ species with matched adult body mass in the 413-tip Burgio et al. 2019 supertree), sample-size-adjusted maximum longevity records from AnAge and the Amniote Life History Database ($n = 152$ adjusted, 171 unadjusted), and endocranial/brain mass ($n = 186$). Models were cross-checked against the phylogenomic tree of Smith et al. (2024; 399 sequence-backed species). Phylogenetic generalized least squares (PGLS, Pagel's λ implemented in Python and validated to $< 10^{-4}$ against `nlme::gls` and `phylolm` in R) quantified allometric slopes and trait residuals. Evolutionary regime shifts across the supertree were inferred with scalar Ornstein-Uhlenbeck expectation-maximization (`PhylogeneticEM`, $K = 0$–12, LINselect criterion). Statistical power was simulated under bivariate Brownian motion across the $n = 105$ genome-matched species set.

To determine whether elevated cancer vulnerability characterizes Psittaculidae broadly or *M. undulatus* specifically, we analyzed two independent veterinary pathology registries: (1) AZA zoological necropsies (Kapsetaki et al. 2024; 5,729 necropsies across 108 bird species including 15 parrots) tested with beta-binomial models and Fisher's exact tests, and (2) the UC Davis / Drury R. Reavill clinical pathology database (10,152 parrot necropsies with ≥ 3 examined tissues), applying direct age standardization to rule out captive age-structure confounding.

2.2 Phase 1: Systematic Audit of 126 Genome Assemblies

All 126 NCBI assemblies matching species with trait data (96 Iridian short-read Illumina/SPAdes drafts + 30 chromosome-scale or long-read genomes from VGP, B10K, Wellcome Sanger, and BGI) were downloaded on-the-fly, evaluated, and removed to minimize storage footprint. Gene completeness was measured via `compleasm` v0.2.10 against `aves_odb10` (8,338 orthologs). To detect sample swaps and library cross-contamination, mitogenome contigs identified via *COX1/CYTB/ND2* from five divergent parrots were queried against 6,700 GenBank avian vouchers strictly *excluding* Iridian/Hains/Pirro records. Nuclear distances across 176 assemblies (96 Iridian + 48 independent replicates + 32 putative donor species) were computed using Mash v2.3 ($k = 21$, 50,000 hashes). Pre-registered inclusion rules required $\text{BUSCO}_{\text{complete}} \geq 70\%$, passing nuclear Mash placement (residual ≤ 0.006 at N50 ≥ 5 kb), and deduplication to one best assembly per species, yielding 71 validated species (58 frozen discovery on Oct. 28, 2026 + 13 newly finalized holdouts). Quality bias was tested via PGLS of residual longevity against BUSCO completeness, $\log_{10}$(contig N50), GC%, and assembly length deviation.

2.3 Phase 2: Proteome Extraction, H8 Panel, Null Pool, SRA Validation, and 71-Species Extension

All 16,045 canonical *M. undulatus* RefSeq proteins were aligned against every approved genome via `miniprot` v0.18 (`-outn=3`, scoring ORFs as *complete* when coverage $\geq 90\%$, identity $\geq 60\%$, and free of frameshifts or premature stops). In the 58-species discovery panel (15 Psittaculidae, 43 non-Psittaculidae), codon alignments (`MAFFT` + `PAL2NAL`) were constructed for 101 candidate genes partitioned across eight pre-specified modules: M1 (p53/cell cycle), M2 (PI3K/mTOR/proliferation), M3 (apoptosis), M4 (DNA DSB/BER repair), M5 (NRF2/xenobiotics), M6 (antioxidants), M7 (telomere/shelterin), and M8 (proteostasis/autophagy). We evaluated relaxation/intensification in Psittaculidae (stem + internal branches as `TEST`) with `HyPhy RELAX` (`--models Minimal`) and episodic positive selection on the Psittaculidae stem and *M. undulatus* tip branch with `HyPhy aBSREL` across both topologies. Benjamini-Hochberg FDR was applied across all testable panel genes per tree (86–87 genes) on top of HyPhy's per-gene Holm-Bonferroni correction across the two test branches. Modular statistics (proportion relaxed at $q < 0.10, k < 1$ and mean $\log_2 k$) were benchmarked against 10,000 module-label permutations and 1,000 length- and background-rate-matched pseudo-panels sampled from 298 genomic control genes. Disrupted calls were verified against a second long-lived reference proteome (*Strigops habroptila*) and raw Illumina SRA reads (`bwa-mem2`/`samtools`). Upon finishing Phase 1, all 97 alignable H8 panel genes were also extended to the full 71-species cohort (`results/phase2_genes_71sp/`) with confirmatory `RELAX` and `aBSREL` verification on key H8 genes.

2.4 Phase 3: Genome-Wide `RERconverge`, 3D Mapping, and 9-Point Red-Team Audit

Across 7,756 coding genes with intact ORFs in $\geq 50\%$ of the 58 discovery species, amino-acid branch lengths were optimized under LG (`phangorn::optim.pml`)

on both topologies. Relative evolutionary rates (`RERconverge` v0.3.0, `getAllResiduals`, square-root transform, weighted, column-scaled) were correlated (`winsorize = 3`) with `rel_log_life_expectancy` ($n = 58$), `rel_log_brain` ($n = 50$), and `rel_brain_given_le` ($n = 50$), plus within-Psittaculidae ($n = 15$) and within-rest ($n = 43$) subsets, using 1,000 topology-conditioned Brownian motion permulations (`Rcpp fastCorPerms`). Enrichment was tested over 1,780 GO Biological Process terms (≥ 10 genes, `fastWilcoxGMTAll`) and six predeclared exploratory functional axes. AlphaFold structures for `POT1`, `RTEL1`, `PTEN`, and `MAD2L2` were segmented by Predicted Aligned Error (PAE) to map lineage-specific substitutions.

Finally, nine adversarial Red-Team audits tested: (1) seed reproducibility ('seed=20261028' vs '999999'), (2) negative control traits and 200-replicate Brownian null distributions for genome-wide coupling r, (3) technical confounders (length, tree length, GC3, missingness, gaps) and zero-branch invariance, (4) leave-one-clade-out perturbations, (5) true tip-OLS, branch-path, and analytical partial orthogonalization of `rel_brain_given_le`, (6) empirical `aBSREL` base rates across control genes, (7) `MAFFT L-INS-i`, stringent gap/cluster window masking, `Double+Triple` multi-hit models, and kakapo dual-reference coordinate concordance in telomeric genes, (8) 71-species panel / 13-species holdout and 30-split cross-validation replication, and (9) an exhaustive statistical inventory of all 78,802 tests and null outcomes.

3 Results

3.1 Macroevolution of Parrot Longevity and Necropsy Epidemiology (Phases 0 & 0b)

Across 218 parrots, PGLS with Pagel's $\lambda = 0.65$ outperforms Brownian (ΔAIC = 69.9) and single-optimum OU (ΔAIC = 14.9) models, yielding an allometric slope of 0.349 ± 0.032 (**Fig. 1**, **Table 1**). Ancestral reconstruction under scOU (`PhylogeneticEM`, **Fig. 2**, right) recovers an ancestrally long-lived regime (root optimum +0.22) sustained across Neotropical Arini (*Psittacara*, *Ara*, *Anodorhynchus*), Androglossini (*Amazona*), Cacatuidae, and Strigopidae, punctuated by three rate reductions led by the ancestral branch of Psittaculidae ($\Delta\theta = -0.35$, $n = 101$ descendant tips; direct PGLS indicator $-31\%, p = 0.003$). Crucially, switching to sample-size-corrected maximum longevity ($n = 152$, $\rho = 0.35$ against life expectancy; **Fig. 2**, left) preserves only the Psittaculidae contrast ($-18\%, p = 0.03$, though $K = 0$ in scOU), and re-testing on the dated Smith et al. (2024) molecular tree elevates phylogenetic signal to $\lambda = 0.88$ while weakening the Psittaculidae longevity contrast to borderline significance ($\beta = -0.076 \pm 0.039$, -25%, $p = 0.050$–0.052). Thus, clade-level longevity reduction in Psittaculidae represents a single topology-sensitive event.

Conversely, veterinary neoplasia epidemiology provides an unambiguous, topology-independent signal: *Melopsittacus undulatus* is an extreme tumor outlier relative to all other parrots including fellow Psittaculidae lories (**Table 1**). In AZA records ($n = 5,729$ necropsies), budgerigars display 20.8% neoplasia (17.4% malignant) versus 1.5% (0.8% malignant) in lories (OR = $17, p \approx 10^{-21}$; beta-binomial outlier $p = 0.002$). In the independent UC Davis database ($n = 10,152$ necropsies), age-standardized tumor prevalence in *M. undulatus* reaches 25.9% versus 5.4% in lories ($4.8\times, p \approx 10^{-16}$) and 9.1% in cockatiels (*Nymphicus hollandicus*, $p \approx 10^{-17}$), ruling out domestication or captive age structure as sole explanations.

Table 1: Phase 0/0b macroevolutionary allometry, topology sensitivity, and veterinary necropsy epidemiology.

Analysis / Dataset	Model / Contrast	Estimate	p / ΔAIC
PGLS LE ~ Mass ($n = 218$)	Pagel's $\lambda = 0.65$ (slope)	$+0.349 \pm 0.032$	$\Delta\text{AIC} = 0$
scOU Regime Shift (`Burgio`)	Ancestral Psittaculidae ($K = 3$)	-0.35 (vs. $+0.22$)	LINselect
Psittaculidae LE Contrast	`Burgio 2019` / `Smith 2024`	-31% / -25%	0.003 / 0.050
Max Longevity Adj. ($n = 152$)	Concordance with Rel. LE	Spearman $\rho = 0.35$	$p = 0.03$ (Psl)
AZA Neoplasia ($n = 5,729$)	*M. undulatus* vs. Lories	20.8% vs. 1.5%	OR=17, 10^{-21}
UC Davis Age-Std ($n = 10,152$)	*M. undulatus* vs. Lories / Cockatiel	25.9% vs. 5.4% / 9.1%	10^{-16} / 10^{-17}

3.2 Phase 1: Assembly Audit Uncovers Cross-Contamination and Yields 71 Species

Completing `compleasm aves_odb10`, mitogenome BLAST, and Mash nuclear audits across all 126 assemblies (**Table 2**; **Fig. 3**; Table S1) exposed pervasive batch quality failures in short-read drafts. Across 96 Iridian assemblies, median completeness was 76.1%, and 0/31 assemblies with contig N50 < 5 kb passed the 70% cutoff. Critically, 61/96 Iridian genomes contained complete foreign mitochondrial genomes (88% matching donors sequenced in the same Iridian batch, including reciprocal lab swaps), and four assemblies failed nuclear Mash identity: "*Agapornis nigrigenis*" (`GCA_035041115.1`) is a passerine (*Garrulax*, nuclear Mash ≥ 0.145), "*Aratinga jandaya*" (`GCA_026284485.1`) is *Poicephalus meyeri* (Mash = 0.0039), and two assemblies (*Cacatua haematuropygia*, *Trichoglossus haematodus*) are chimeras contaminated with the same passerine DNA. Deduplicating and applying filters ($\text{BUSCO}_{\text{complete}} \geq 70\%$) finalized 71 validated species (58 discovery + 13 holdout; Table S1), with zero technical correlation between assembly quality ('busco_C' $p = 0.111$–0.352, 'contig N50' $p = 0.198$–0.249, GC $p \geq 0.097$) and residual longevity.

Table 2: Summary of the 126-assembly genomic audit (`species_genomes.tsv`).

Audit Category / Quality Filter	Assemblies	Species
Total audited (`compleasm aves_odb10`)	126	105
Iridian short-read batch / Independent long-read	96 / 30	96 / 29
Excluded for $\text{BUSCO}_{\text{complete}} < 70\%$	39	32
Excluded for nuclear swap / chimera (Mash)	4	2
Included species (BUSCO $\geq 70\%$)	**71**	**71**
Psittaculidae / Other Psittaciformes	20 / 51	20 / 51
Strict original criterion ($\geq 85\%$ & N50 ≥ 20 kb)	33	33
Frozen discovery panel (Phases 2–3) / Holdout set	58 / 13	58 / 13

3.3 Phase 2: Refutation of Tumor-Suppressor Loss, Telomere Signal, 71-Species Confirmation, and Multi-Hit Collapse

Across the 87 alignable H8 panel genes in 58 species (97 aligned across 71 species; **Fig. 4**), we reject modular loss or relaxation of cancer-defense networks:

1. **Zero confirmed pseudogenizations:** All 329 apparent disruptions (451 in 71 species; Table S2) resolved as single-reference alignment artifacts (`OGG1`, `MUTYH`, `MYC`, `HMOX1`), heterozygous stop variants (`PSMD6` in *Loriculus galgulus*: 21 C vs. 20 T SRA reads), or 0× Illumina dropout across a 75–77% GC exon 1 in `ACD` (*Psittacus erithacus*). Because GC-rich `TP53` and `RB1` remained fragmented in short-read drafts (intact in only 4 and 19 of 71 species), putative loss in those two genes remains untestable outside long-read assemblies.

2. **Modular null benchmark:** Neither module heterogeneity in mean $\log_2 k$ ($p = 0.164$ Burgio, 0.955 Smith) nor any module M1–M8 departed significantly from 1,000 matched pseudo-panels sampled from 298 genomic controls (lowest nominal M2 $p = 0.024$ was driven by two $k \approx 0$ boundary genes and collapsed to $p = 0.070$ under medians; Table S3).

3. **Telomeric (M7) episodic signal vs. Red-Team controls and 71-species confirmation:** Under standard `aBSREL`, 5/8 (62.5%) telomeric M7 genes (`RTEL1` $p = 1.73 \times 10^{-10}$, `TERF2` 1.79×10^{-6}, `TERT` 2.63×10^{-4}, `ACD` 2.46×10^{-4}, `TERF1` 3.98×10^{-4}) and `MYC` (1.69×10^{-8}) reached $q < 0.10$ on *M. undulatus* (BH-corrected across all 86 tested panel genes), contrasting starkly with non-M7 panel modules (1/78 = 1.3%) and 19 control genes (1/19 = 5.3% nominal; **0/19 FDR**; Fisher $p = 0.00069$; 0/298 controls had $\omega > 1$ in RELAX). In Psittaculidae, non-boundary RELAX intensification replicated in `TERT` ($k = 5.51/3.10$, $q = 0.0008/0.011$) and `MRE11` ($k = 2.60/2.61$, $q = 0.002/0.005$), with moderate relaxation in `DKC1` ($k = 0.49$) and `NQO1` ($k = 0.34$). Expanding alignments up to 71 species (**Table S2**; nearly doubling taxon sampling for `TERT` 15 → 34, `RTEL1` 23 → 35, `TERF2` 18 → 29, `POT1` 8 → 26, and `TERF1` 9 → 25) sharpens the telomeric cluster while falsifying all non-telomeric RELAX hits except `NQO1`: `TERT` ($k = 2.24$, RELAX $p = 1.32 \times 10^{-7}$; `aBSREL` $p = 1.04 \times 10^{-4}$), `POT1` (k strengthens from $1.94, p = 0.075$ to $8.17, p = 2.27 \times 10^{-5}$), `DKC1` ($k = 0.55, p = 7.15 \times 10^{-6}$), `RTEL1` (moves off the $k = 0$ boundary to $k = 0.52, p = 0.0159$; `aBSREL` $p = 9.43 \times 10^{-11}$), `TERF2` ($k = 0.59, p = 0.0161$; `aBSREL` $p = 5.95 \times 10^{-5}$), and `NQO1` ($k = 0.71, p = 0.0091$) preserve their selective regimes, whereas `MRE11` (49 → 63 spp) collapses to non-significance ($k = 0.86, p = 0.106$), `SOD1` flips from apparent intensification ($k = 31.65$) to a zero boundary ($k = 3.8 \times 10^{-6}$), and `CCNE1` reverses and becomes null ($k = 1.76, p = 0.229$).

   Crucially, however, Red-Team alignment auditing (**Table 3**) demonstrates that although *S. habroptila* dual-reference coordinates verify 100% single-locus orthology for `RTEL1` (40/40), `TERF2` (30/30), `TERT` (35/35), `TERF1` (27/27), and `POT1` (27/27) and `MAFFT L-INS-i` reproduces every signal, **cleaning gappy/noisy windows** (> 15% gap columns and 10-codon windows with ≥ 3 AA differences) extinguishes `RTEL1` ($p = 1.00$), `ACD` (0.125), and `TERF2` (0.068), leaving only `TERT` ($p = 0.0052$) and `TERF1` ($p = 0.0037$). Further allowing simultaneous multinucleotide changes (`--multiple-hits Double+Triple`) collapses all five signals ($p = 0.967$–1.00; `ACD` 0.050–0.132).

3.4 AlphaFold 3D Structural Mapping

Projecting alignment substitutions onto PAE-partitioned AlphaFold models (**Fig. 5**; Table S6) separates C-terminal/surface remodeling from ultra-conserved enzymatic pockets. In `RTEL1` (1,219 AA, global pLDDT = 71.67), 43 budgerigar-specific changes fall within the N-terminal Rad3/FeS/Arch helicase core (residues 1–758) whereas 115 concentrate in the C-terminal Harmonin-N and PIP/RING region (893–1,219). In `POT1` (780 AA, global pLDDT = 79.21), 34 substitutions map across OB1/OB2 and 16 across TPP1-binding OB3/HJRL (mean mutated-site pLDDT = 72.6). By contrast, `PTEN` (403 AA, pLDDT = 82.47) is 100% invariant across its catalytic P-loop (`H123-R130`, pLDDT = 93.0) and Phosphatase+C2 superdomain (14–350), with polymorphisms restricted to its disordered C-tail (351–403, pLDDT = 35.1), matching `MAD2L2`'s rigidly conserved HORMA core (pLDDT = 90.39).

3.5 Phase 3 (`RERconverge`): Genome Coupling, Orthogonalization, and Zero-Branch Artifacts

Across 7,756 genes in 58 species (**Fig. 6**), 190–277 genes per trait passed parametric $q < 0.10$, yet zero reached genome-wide permulation FDR (permQ < 0.10) owing to phylogenetic conservatism and the 1,000-permulation resolution floor ($\text{permP}_{\text{min}} = 0.001996$). Among 1,780 GO terms, nominal directional consistency across both trees emerged in DNA recombination ($p = 3.2 \times 10^{-4}$), DNA damage response ($p = 0.0076$), telomere maintenance ($p = 0.013$), mitochondrial translation ($p = 0.0021$), macroautophagy ($p = 0.030$), and synaptic vesicle recycling ($p = 0.0030$), and among six exploratory functional axes only Neuroprotection & Proteostasis shifted above background in both traits ($p = 0.014$–0.046).

Unadjusted trait-RER correlations across all 7,756 genes correlate strongly ($r = 0.666$ Burgio, 0.667 Smith, 0.748 on the matched 50-species matrix), are 100% deterministic ($\max|\Delta\text{Rho}| = 5 \times 10^{-16}$ across seeds), and persist after removing 14 linear, log, and quadratic technical confounders ($r = 0.653$; 0.630–0.696 across quar-

Table 3: Red-Team sensitivity audit of telomeric (M7) `aBSREL` signals on the *M. undulatus* branch and dual-reference orthology verification (`7_telomere_alignment_and_rbh.tsv`).

Gene	Spp.	*M. und.* AA Subs (Isolated / Noisy)	Orig. MAFFT p	L-INS-i p	Window-Cleaned p	Multi-Hit (`Double+Triple`) p	Kakapo Orthology
`RTEL1`	23	279 (58 / 221)	1.73×10^{-10}	1.28×10^{-9}	**1.0000**	**0.9670**	40/40 (2 loci *Melo*)
`TERF2`	18	89 (32 / 57)	1.79×10^{-6}	2.44×10^{-6}	**0.0677**	**1.0000**	30/30
`TERT`	15	196 (95 / 101)	2.63×10^{-4}	1.12×10^{-4}	**0.0052**	**1.0000**	35/35
`ACD`	6	102 (42 / 60)	2.46×10^{-4}	1.23×10^{-4}	**0.1248**	**0.0502** (0.1320 cl.)	8/11 (2 loci *Melo*)
`TERF1`	9	23 (15 / 8)	3.98×10^{-4}	1.99×10^{-4}	**0.0037**	**1.0000** (0.3151 cl.)	27/27
`POT1`	8	84 (52 / 32)	1.0000	1.0000	1.0000	1.0000	27/27

tiles). Nevertheless, our Red-Team audit (**Table 4**; Tables S4–S5) exposes three essential caveats:

1. **True partial orthogonalization halves the residual brain signal:** Subtracting an unwhitened PGLS slope ($\hat{\beta}_{\text{GLS}} = 0.040, \lambda = 0.90$) left `rel_brain_given_le` correlated with longevity at $r_{\text{tip}} = 0.427$ ($\text{cor}(\text{Rho}_{\text{Brain}}, \text{Rho}_{\text{PGLS}}) = 0.999$), explaining why r fell only from 0.666 to 0.639. Applying true tip-OLS orthogonalization ($\hat{\beta}_{\text{OLS}} = 0.258, r_{\text{tip}} = 0$) cuts r in half to **0.332** (0.369 in 50sp), and branch-path orthogonalization ($r_{\text{path}} = 0$) gives **0.389**. Furthermore, Brownian permulations yield a wide null SD (0.29–0.35, max +0.716, empirical $p = 0.010$–0.025) because genome-wide branch-rate covariance amplifies any shared branch-path correlation into r_{Rho} ($\text{cor}(r_{\text{Rho}}, r_{\text{path}}) = 0.883$–$0.913$), whereas mass-orthogonalized clutch size drops to the noise floor ($r = -0.102$).

2. **Clade dependence:** Dropping any single clade leaves $r_{\text{Rho}} \in [0.649, 0.798]$, but removing all 13 Psittaculidae drops phenotypic r_{tip} from 0.478 to **0.195** (Spearman **0.097**) and r_{Rho} from 0.748 to **0.522**.

3. **Zero-branch invariance drives 89.3% of shared single-gene hits:** Total tree length negatively correlates with signed $\text{Rho}_{\text{Brain}}$ ($r = -0.178, p = 1.8 \times 10^{-56}$). Crucially, **117/131 (89.3%)** shared significant hits (**Tier A**: `PTEN`, `MAD2L2`, `HSPA8`, `CTNNB1`, `SOX14`, `RTN1`, `PSMA4`) possess **97–99% zero-length branches** ($\leq 10^{-6}$). Their studentized residuals mirror $\pm\sqrt{\text{nv}}$ from the genome background average—accounting for why `PTEN` flips from −0.374 to +0.184 when normalized within a small 91-gene panel yet holds 100% sign concordance across 30 within-cohort CV splits sharing `nv`. Restricting to **Tier B informative genes** (≥ 20 nonzero branches, $n = 6,182$) retains 14 shared hits (**Table 5**; Table S5), led by `TKT`, `PHOSPHO1`, `TADA2A`, `SLC4A7`, and `HUS1`.

Table 4: Red-Team stress tests on Phase 3 genome-wide coupling $r(\text{Rho}_{\text{LE}}, \text{Rho}_{\text{Brain}})$.

Audit Test / Perturbation	r_{tip}	r_{path}	$r(\text{Rho}_1, \text{Rho}_2)$
Original (`seed=20261028`) / `seed=999999`	+0.478	+0.415	**0.6663 / 0.6663**
50-species matched `rers_50` matrix	+0.478	+0.425	**0.7478**
Adjusted for 14 quadratic+log confounders	–	–	**0.6532**
Negative control: mass-resid. clutch size ($n = 51$)	-0.173	-0.214	**-0.1024**
200 BM permulations (independent pairs)	~ 0	-0.002	SD=0.289, $p = 0.010$
Observed LE × 200 Brain permulations	~ 0	+0.033	SD=0.350, $p = 0.025$
Leave-one-clade-out: Drop *M. undulatus* (-1)	+0.446	+0.405	0.7285
Leave-one-clade-out: Drop Arini (-12)	+0.380	+0.241	0.6489
Leave-one-clade-out: **Drop Psittaculidae (-13)**	**+0.195**	**+0.209**	**0.5225**
Unwhitened PGLS `rel_brain_given_le`	+0.427	+0.370	0.6385
True Tip-OLS orthogonalization ($r_{\text{tip}} = 0$)	**0.000**	-0.063	**0.3322**
True Branch-path orthogonalization ($r_{\text{path}} = 0$)	–	**0.000**	**0.3888**
Analytical per-gene partial correlation	–	0.000	**0.4135**

Table 5: Selected Tier A (ultra-conserved > 97% zero branches) vs. top Tier B (≥ 20 nonzero branches) shared candidates ($P < 0.05$ & permP < 0.05 for both longevity and brain; full list in Table S5).

Gene	Tier (% Zero Br.)	Rho_{LE}	permP_{LE}	$\text{Rho}_{\text{Brain}}$	$\text{permP}_{\text{Brain}}$
`PTEN`	Tier A (99.1%, 1 nz)	-0.374	0.0080	-0.424	0.010
`MAD2L2`	Tier A (99.0%, 1 nz)	+0.418	0.0120	+0.534	0.002
`HSPA8`	Tier A (98.2%, 2 nz)	+0.404	0.0080	+0.354	0.030
`CTNNB1`	Tier A (99.1%, 1 nz)	-0.439	0.0040	-0.459	0.012
`TKT`	Tier B (60.9%, 27 nz)	+0.408	0.0040	+0.347	0.032
`PHOSPHO1`	Tier B (52.3%, 53 nz)	-0.261	0.0120	-0.267	0.024
`TADA2A`	Tier B (72.6%, 20 nz)	+0.316	0.0220	+0.334	0.012
`SLC4A7`	Tier B (44.8%, 37 nz)	+0.296	0.0319	+0.304	0.046
`HUS1`	Tier B (54.1%, 51 nz)	-0.222	0.0120	-0.253	0.018
`DEPDC4`	Tier B (21.5%, 73 nz)	-0.250	0.0100	-0.214	0.034

4 Discussion

Our integrated audit reconciles biological signal with methodological artifacts across three levels of avian comparative genomics:

1. **Genome auditing is non-negotiable for public short-read drafts:** The prevalence of foreign mitochondrial genomes (61/96) and passerine contamination/chimerism (3/96) in batch-deposited genomes demonstrates that reliance on NCBI metadata without nuclear Mash and multi-reference exon validation will generate spurious claims of gene loss and episodic selection.

2. **Budgerigar telomere divergence involves clustered/multinucleotide dynamics rather than gene loss:** Short-lived parrots retain intact DNA-repair and checkpoint toolkits. While M7 telomeric genes (`TERT`, `TERF1`, `RTEL1`, `TERF2`, `ACD`) show true orthology with *Strigops* and exceed empirical control rates under standard `aBSREL`, sensitivity to noisy-window masking and complete absorption under `Double+Triple` multi-hit models suggest clustered multinucleotide mutational regimes or alignment-local divergence in C-terminal `RTEL1` and OB-domain `POT1` rather than classic dispersed positive selection. Targeted long-read sequencing and functional telomere-length assays in Psittaculidae remain key priorities, alongside resolving GC-rich `TP53` and `RB1` in chromosome-scale genomes.

3. **Conserved proteomes challenge branch-rate correlation interpretation:** Even after proper OLS/branch-path orthogonalization ($r = 0.33$–0.39), brain size and lifespan remain partially coupled through shared rate structure on the parrot tree—anchored heavily by Psittaculidae versus long-lived radiations. Critically, future `RERconverge` applications on deeply conserved vertebrate pro-

teomes must filter invariant genes ($> 80\%$ zero-length branches), where studentized residuals mirror $-\hat{b}_g\sqrt{\mathrm{nv}}$ rather than gene-specific convergent substitutions.

Data & Dashboard Availability

Full scripts, confirmatory 71-species outputs, Supplementary Materials (`supplementary_parrots_EN.pdf`, Tables S1–S6), Red-Team audit logs, and an interactive offline HTML raw-data explorer (`parrots_raw_data_dashboard.html`) are hosted at `github.com/redkk123/Parrots` (branch `claude/new-session-lwh7xh`).

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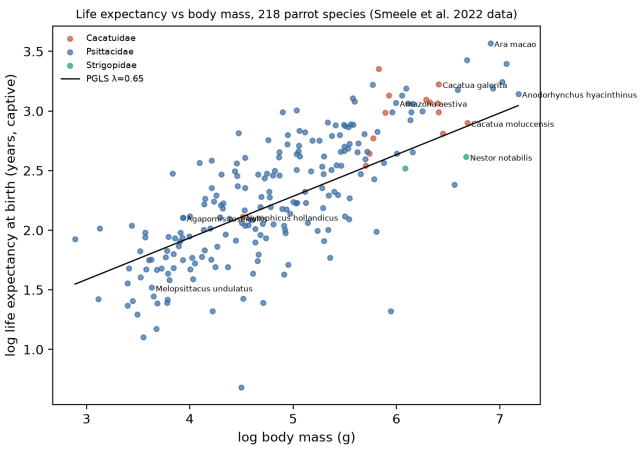
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[5] Wertheim, J. O. et al. (2015). RELAX: detecting relaxed selection in a phylogenetic framework. *Molecular Biology and Evolution*, 32(3), 820–832.

[6] Wirthlin, M. et al. (2018). Parrot genomes and the evolution of heightened longevity and cognition. *Current Biology*, 28(24), 4001–4008.


Relative life expectancy mapped on the tree (exploratory BM reconstruction)
● chromosome-scale genome, ○ draft/in progress, ◌ reported (level unknown)
Melopsittacus undulatus ●
Pseudeos fuscata
Oreopsittacus arfaki
Charmosyna josefinae
Charmosyna stellae
Charmosyna placentis
Charmosyna papou
Charmosyna pulchella
Phigys solitarius
Vini australis
Vini peruviana
Lorius chlorocercus
Lorius lory
Lorius garrulus
Lorius domicella
Glossopsitta concinna
Glossopsitta porphyrocephala
Glossopsitta pusilla
Psitteuteles versicolor
Chalcopsitta duivenbodei
Chalcopsitta cardinalis
Psitteuteles goldiae ○
Eos squamata
Eos reticulata
Eos cyanogenia
Eos histrio
Eos bornea
Trichoglossus forsteni
Trichoglossus johnstoniae
Trichoglossus rosenbergii
Trichoglossus weberi
Trichoglossus moluccanus
Trichoglossus ecklonii
Trichoglossus capistratus
Trichoglossus chlorolepidotus
Trichoglossus haematodus
Trichoglossus rubritorquis
Psitteuteles iris
Trichoglossus flavoviridis
Trichoglossus ornatus
Chalcopsitta atra
Chalcopsitta scintillata
Neopsittacus musschenbroekii
Neopsittacus pullicauda
Psittaculirostris salvadorii
Psittaculirostris edwardsii
Psittaculirostris desmarestii
Cyclopsitta diophthalma
Agapornis taranta
Agapornis canus
Agapornis roseicollis ○
Agapornis nigrigenis
Agapornis lilianae
Agapornis personatus
Agapornis fischeri
Loriculus vernalis
Loriculus philippensis
Loriculus galgulus
Lathamus discolor ●
Eunymphicus cornutus
Cyanoramphus unicolor
Cyanoramphus novaezelandiae
Cyanoramphus auriceps
Cyanoramphus malherbi
Psephotus haematonotus
Northiella haematogaster
Barnardius zonarius
Platycercus icterotis
Platycercus eximius
Platycercus venustus
Platycercus adscitus
Purpureicephalus spurius
Psephotellus varius
Psephotellus dissimilis
Psephotellus chrysopterygius
Neopsephotus bourkii
Neophema elegans
Neophema chrysostoma
Neophema splendida
Neophema chrysogaster
Neophema pulchella
Psittacula eupatria
Psittacula roseata
Tanygnathus lucionensis
Tanygnathus sumatranus
Tanygnathus megalorynchos
Psittacula krameri ●
Psittacula longicauda
Psittacula alexandri
Psittacula derbiana
Psittacula columboides
Psittinus cyanurus
Psittacula cyanocephala
Psittacula himalayana
Eclectus roratus
Polytelis swainsonii
Alisterus amboinensis
Aprosmictus erythropterus
Polytelis alexandrae
Alisterus chloropterus
Polytelis anthopeplus
Psittrichas fulgidus
Coracopsis vasa
Coracopsis nigra
Pionopsitta pileata
Triclaria malachitacea
Pionus senilis
Pionus menstruus
Pionus sordidus
Pionus fuscus
Pionus maximiliani
Amazona agilis
Amazona vittata ○
Amazona lilacina
Amazona albifrons
Amazona xanthotora
Amazona ventralis
Amazona viridigenalis
Amazona collaria
Amazona leucocephala
Amazona ochrocephala ●
Amazona barbadensis
Amazona aestiva ○
Amazona tucumana
Amazona pretrei
Amazona autumnalis
Amazona dufresniana
Amazona rhodocorytha
Amazona finschi
Amazona brasiliensis
Amazona amazonica
Amazona guildingii
Alipiopsitta xanthops
Brotogeris pyrrhoptera
Brotogeris chrysoptera
Brotogeris cyanoptera
Brotogeris tirica
Brotogeris versicolorus
Myiopsitta monachus
Cyanoliseus patagonus
Psittacara finschi
Psittacara erythrogenys
Eupsittula canicularis
Eupsittula nana
Eupsittula pertinax
Psittacara leucophthalmus
Psittacara wagleri
Psittacara mitratus
Psittacara holochlorus
Psittacara rubritorquis
Orthopsittaca manilatus
Ara rubrogenys
Ara ambiguus
Ara chloropterus
Ara macao ○
Ara ararauna ●
Primolius maracana
Primolius auricollis
Primolius couloni
Aratinga solstitialis
Aratinga nenday
Aratinga jandaya
Aratinga auricapillus
Guaruba guarouba ●
Psittacara acuticaudatus
Diopsittaca nobilis
Diopsittaca cumanensis
Anodorhynchus leari
Anodorhynchus hyacinthinus ○
Ennocognathus leptorhynchus
Rhynchopsitta pachyrhyncha
Pyrrhura picta
Pyrrhura pacifica
Pyrrhura hoffmanni
Pyrrhura emma
Pyrrhura frontalis
Pyrrhura cruentata
Pyrrhura lepida
Pyrrhura griseipectus
Pyrrhura rupicola
Pyrrhura melanura
Pyrrhura rhodocephala
Pyrrhura perlata
Deroptyus accipitrinus
Pionites melanocephalus
Pionites leucogaster
Forpus spengeli
Forpus cyanopygius
Forpus passerinus
Forpus coelestis
Polioptilagon aymara
Polioptilagon australis
Bolborhynchus lineola
Pyrilia cryptoxanthus
Pyrilia ruappelli
Pyrilia gulielmi
Pyrilia robustus
Pyrilia ruffirostris
Pyrilia meyeri
Pyrilia senegalus
Psittacus erithacus
Psittacus timneh
Probosciger aterrimus
Eolophus roseicapilla
Cacatua ducorpsii ○
Cacatua tenuirostris ○
Cacatua haematuropygia
Cacatua sanguinea ○
Cacatua ophthalmica
Cacatua moluccensis
Cacatua alba
Cacatua sulphurea
Cacatua galerita ●
Nymphicus hollandicus
Calyptorhynchus lathami
Zanda funerea
Nestor notabilis ○
Nestor meridionalis
0
10
20
30
40
50
60
Myr from root (Burgio et al. 2019 supertree)
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
relative log life expectancy (BM estimate)


Life expectancy vs body mass, 218 parrot species (Smeele et al. 2022 data)
Cacatuidae
Psittacidae
Strigopidae
PGLS λ=0.65
Ara macao
Cacatua galerita
Amazona aestiva
Anodorhynchus hyacinthinus
Cacatua moluccensis
Nestor notabilis
Agapornis roseicollis
Nymphicus hollandicus
Melopsittacus undulatus
3.5
3.0
2.5
2.0
1.5
1.0
log life expectancy at birth (years, captive)
3
4
5
6
7
log body mass (g)

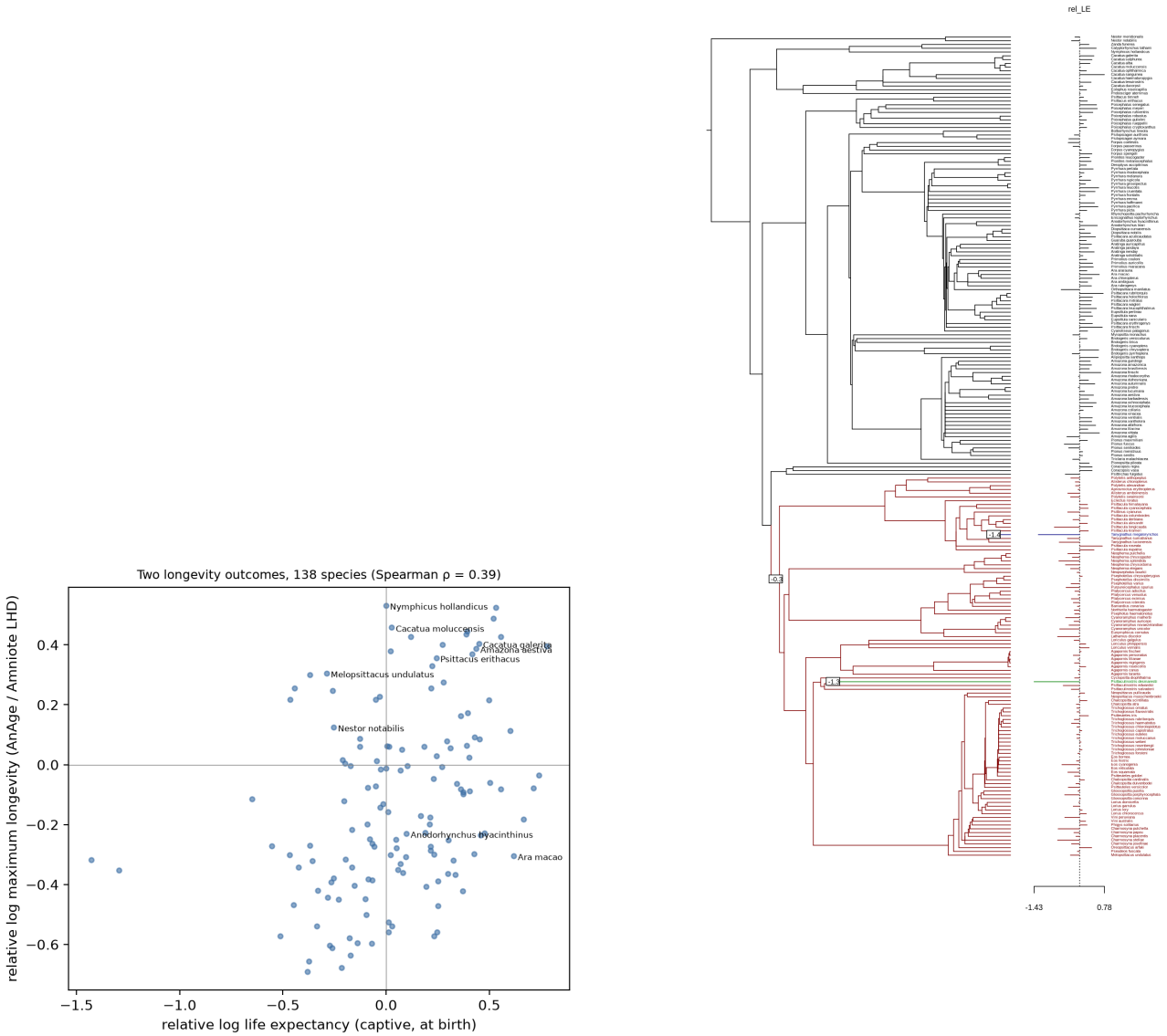


Figure 2: Phase 0b Two-Outcome Sensitivity & Regime Shifts: (**Left**) Modest concordance ($\rho = 0.35$) between relative life expectancy and sample-size-adjusted maximum longevity across 138 parrots, showing outcome sensitivity in kea and macaws. (**Right**) `PhylogeneticEM` scOU inference ($K = 3$ reductions from ancestral optimum +0.22), led by a −0.35 log shift on the Psittaculidae stem ($p = 0.003$ Burgio vs. $p = 0.052$ Smith 2024).

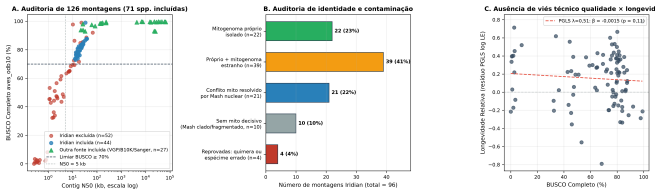


Figure 3: Phase 1 Assembly Audit (126 **assemblies** / 105 **species):** `compleasm aves_odb10` completeness against contig N50, highlighting exclusion of fragmented drafts ($<$ 70%), nuclear swaps/chimeras, and absence of technical bias with relative longevity ($p \geq 0.11$).

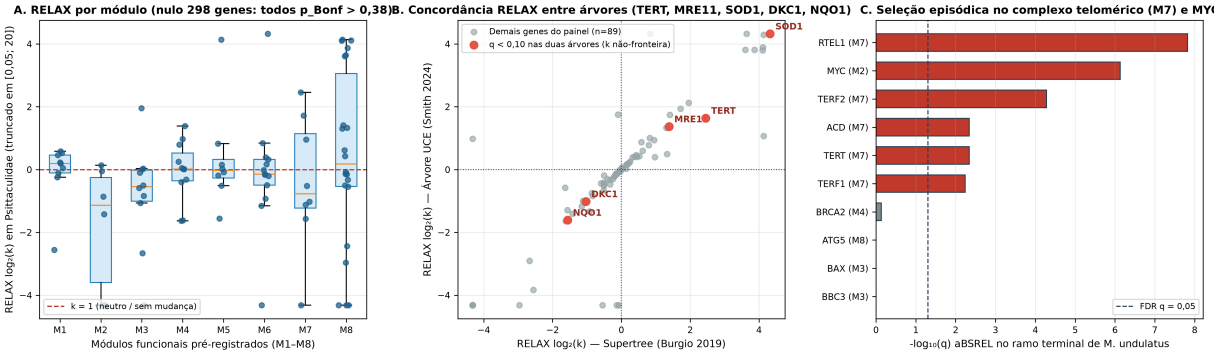


Figure 4: Phase 2 Selection Tests (H8, 58 species): (A) Module $\log_2 k$ distributions vs. 298-gene null model; (B) Per-gene RELAX significance; (C) Concentration of standard `aBSREL` calls in telomere genes (M7) and `MYC` prior to Red-Team window-cleaning and Multi-Hit auditing; (D) Resolution of apparent gene disruptions via second-reference and SRA read checks.

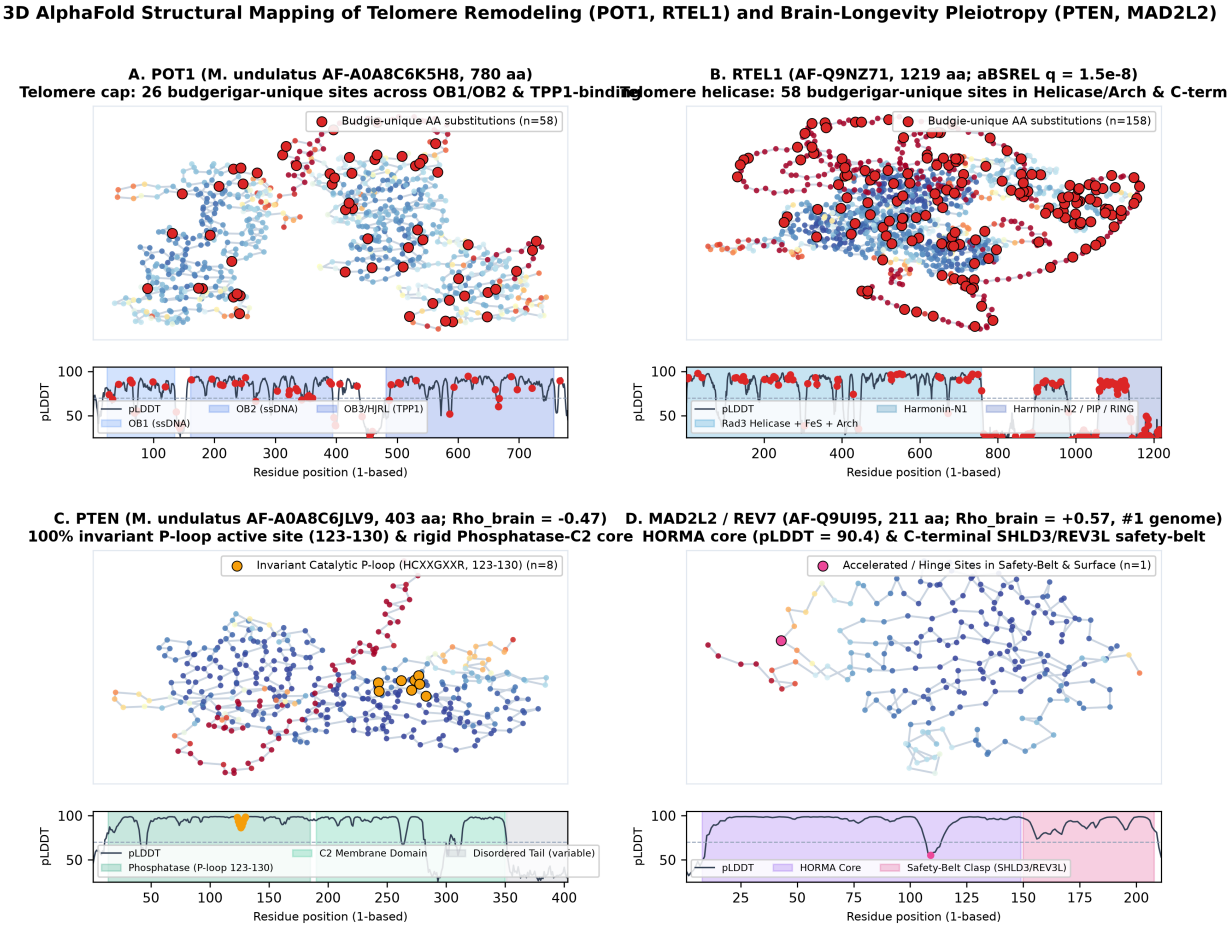


Figure 5: AlphaFold 3D Structural & PAE Domain Mapping (Step 2): Budgerigar substitutions distribute across ssDNA-binding OB1–OB3/HJRL domains in `POT1` (58 sites, mean pLDDT = 72.6) and cluster heavily in the C-terminal Harmonin-N/PIP region of `RTEL1` (115/158 sites), whereas `PTEN` exhibits 100% invariance across its catalytic P-loop (`H123-R130`, pLDDT = 93.0) and Phosphatase+C2 core (residues 14–350).

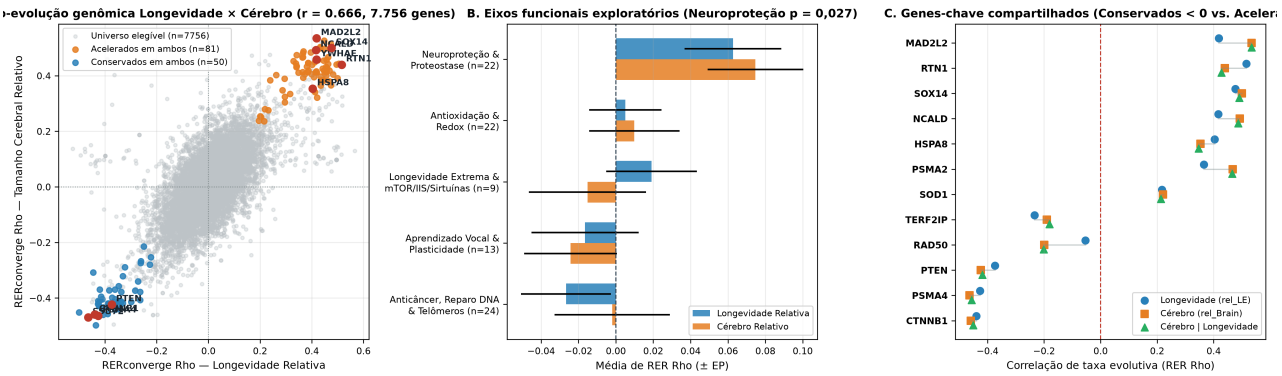


Figure 6: Phase 3 Genome-Wide `RERconverge` Across 7,756 Genes: (A) Unadjusted genome-wide correlation ($r = 0.666$) between Rho_{LE} and $\text{Rho}_{\text{Brain}}$; (B) Shared candidate rankings; (C) Exploratory functional axis shifts highlighting Neuroprotection/Proteostasis ($p = 0.014$–0.046); (D) Within-clade H8 results.